ARC HOBBY · QUICK REFERENCE

Shop tools

Eye protection always. Practice in Arc shows how each tool works; your class teaches its safety rules at the machine, before you use it.

MEASURE AND MARK



Tape measure

Measures lengths; the hook slides so inside and outside measures are both true.

Marking cuts, checking a board's real width.



Speed square

Draws a line square (90°) or at 45° across a board.

Marking cut lines, checking corners, guiding a circular saw.



Clamps

Hold work still, or glued parts tight while the glue dries.

Holding a board to cut or drill it, glue-ups.

Clamp anything you drill with a big bit: it can grab and spin.



Marking gauge

A fence and a sharp pin or wheel that scribes a line a set distance from an edge.

Tenon thickness, dovetail baselines, half-laps.

CUT



Hand saw

Cuts by hand: slow, quiet and safe to learn on.

Small cuts, a few boards.

Clamp it or use a bench hook; start with light pull strokes.



Backsaw (dovetail saw)

A thin saw stiffened along its back, for fine, straight cuts.

Tenon cheeks and shoulders, dovetails, half-laps.

Clamp the work; start with a few light pull strokes on the waste side.



Miter saw `Beginner`

A spinning blade on an arm that cuts a board to length, square or angled.

Boards to length, 45° frame corners.

Hands 6" from the blade. Let it stop before you lift it.



Circular saw `Beginner`

A hand-held blade for straight cuts in boards and sheets.

Plywood sheets, along a straightedge guide.

Two hands on it; support both sides so it doesn't pinch.



Jigsaw `Beginner`

A thin blade that moves up and down to follow curves.

Shaped cuts: a birdhouse gable, a bracket.

Shoe flat on the board; let it stop before lifting it out.



Bandsaw `Intermediate`

A thin blade in a loop, running down through a table.

Curves, cutting circles out roughly, resawing thin boards.

Guard down to just above the wood; push sticks near the blade.



Scroll saw `Beginner`

A fine blade that moves up and down, for tight curves.

Small shapes, letters, inside cut-outs.

Hold-down foot on the wood; fingers to the sides of the blade.



Table saw `Advanced`

A blade up through a table; the board is pushed along a fence.

Rip cuts: making a board narrower.

Guard on, push stick, never stand right behind the blade.

DRILL AND DRIVE



Drill / driver `Beginner`

Drills holes and drives screws; the clutch stops it over-driving.

Pilot holes, screws, dowel and hinge holes.

Clamp the work; sleeves up, hair back.

Twist bits

Round holes up to about ⅜"; a small pilot hole stops the wood splitting.

#8 screw: ³⁄₃₂" pilot in pine, ⅛" in oak, ¹¹⁄₆₄" through.

Countersink bit

Cuts a cone so the screw head sits flush.

Neat screw heads you can sand over.



Spade bit `Beginner`

A flat bit for holes ½" to 1½".

A birdhouse entrance hole.

Clamp it; drill from both sides so it doesn't splinter.



Hole saw

A ring of teeth for holes bigger than 1½".

Pipes, a big birdhouse, a speaker.

Clamp it; slow speed, both hands.

JOIN



Pocket-hole jig `Beginner`

Guides the drill at 15° so a screw joins boards from the hidden side.

Table aprons, face frames, cabinets.



Doweling jig

Lines up holes in two boards for wooden pins.

Hidden joints, edge-gluing boards.



Biscuit joiner `Intermediate`

Plunges a curved slot for a glued wooden biscuit.

Lining up edge-glued panels and cabinet corners.

Clamp the work; both hands on the joiner.



Router `Advanced`

A fast-spinning bit that cuts grooves and shapes edges.

Dados, rabbets, rounded edges.

Clamp the work; against the spin; both hands.



Router table `Advanced`

A router upside down under a table: the wood is pushed past the bit along a fence.

Edge profiles, grooves, tongues, sliding dovetails.

Feed right to left, against the spin; push block and featherboards.



Circle jig `Advanced`

A router on an arm that swings round a pin, cutting a true circle.

Round tabletops, clock faces, discs.

Screw the blank to scrap underneath; several shallow passes.

Chisel

A sharp blade, pushed or tapped, to pare wood away.

Cleaning a dado, squaring a corner, fitting a hinge.

Cut away from your body and your other hand.



Wood glue

Stronger than the wood once clamped and dry.

About an hour in clamps; full strength in a day.

MILL, SHAPE AND FINISH



Jointer `Advanced`

Flattens one face, then squares an edge against its fence.

The first step in milling rough lumber.

Push blocks and the guard always; never under 12".



Thickness planer `Intermediate`

Makes the second face parallel, to an even thickness.

Rough 4/4 lumber to ¾", about ¹⁄₃₂" a pass.

Never under 12"; with the grain; hands clear of the infeed.



Hand plane `Beginner`

Each push shaves a thin curl off the wood.

Smoothing an edge, fitting a drawer.



Wood lathe `Advanced`

Spins the wood while you shape it with chisels on a rest.

Legs, spindles, knobs, finials, bowls.

Face shield; rest close; spin by hand; slow for a square blank.



Random-orbit sander `Beginner`

Sands smooth without swirls, by grit.

120 to smooth, 220 to finish.

Dust mask; let it stop before you put it down.



Sandpaper

Lower numbers are coarser: work up through the grits.

By hand, easing edges, small parts.



Brush and finish

Oil, stain, paint or polyurethane to protect the wood.

Two thin coats beat one thick one.

Oily rags can catch fire: lay them flat outside to dry.

HANG IT, AND PROTECT YOURSELF



Stud finder

Finds the wood studs in a wall, about 16" apart.

Hanging a shelf so the screws go into wood.

Level

The bubble sits between the lines when it's level.

Hanging a shelf straight, checking a top.



Push stick and push block

Keep your hands clear of the blade and the cutterhead.

Table saw rips, jointer passes.



Glasses and ear protection

Keep chips out of your eyes and save your hearing.

Every power tool, every time.

METAL SHOP: CUT, GRIND AND WELD STEEL



MIG welder `intermediate`

Feeds wire through the gun and melts the two pieces together under a cloud of gas.

Welding steel frames: tack first, then short welds.

Helmet down before the trigger, say "covering"; never weld galvanized, painted or oily steel.



Welding gear

Auto-darkening helmet (shade 10–13 for MIG), leather gloves, a flame-resistant jacket.

Every weld. Safety glasses stay on under the helmet.

Cotton or wool under it, never polyester: it melts onto skin.



Fume extractor

Pulls the welding smoke away at the joint.

On and close to the weld, every time.

Keep your head out of the smoke as it rises.



Horizontal bandsaw

Cuts steel to length, square or at 45°, on its own once it's started.

Tube, angle and bar for a frame.

Steel clamped in the vise, long lengths held up, hands clear of the blade.



Metal chop saw

An abrasive wheel or a cold-saw blade that cuts steel fast.

Cutting steel to length when there's no bandsaw.

Steel in the vise, never held; face shield; sparks away from anything that burns.



Angle grinder `intermediate`

Grinds mill scale off, deburrs and smooths welds; cuts with a thin disc.

Cleaning joints before welding, shaping coped ends.

Guard and side handle on, face shield, both hands, sparks away from people.



Magnets, clamps, soapstone

Welding magnets hold pieces at 90° or 45°; locking clamps hold them to the table; soapstone marks steel.

Fitting and tacking a frame square.



Fire extinguisher and fire watch

An ABC extinguisher in reach, and someone watching for sparks.

Every cut, grind and weld; weld in the metal shop, never the wood shop.

Clear 35 ft (11 m) of anything that burns; watch 30 minutes after the last weld.

METAL SHOP: BEND, FOLD, AND THE OTHER WELDERS



Flux-core welder `intermediate`

A wire welder with the flux inside the wire: no gas bottle, so it works outdoors in a breeze.

Thicker steel on a 120 V outlet, outside work. Most MIG welders run it too.

Far more smoke: the extractor at the weld. Chip the slag with glasses on.



Stick welder `intermediate`

An arc from a flux-coated rod melts the rod and the steel together; the rod burns shorter as you weld.

Thick, rusty or outdoor steel. Not thin tube: it burns through.

Dry gloves; never touch the rod and the work at once; hot stubs into a steel can.



TIG welder `advanced`

A tungsten arc melts a puddle; you dab filler rod in by hand, the heat on a foot pedal, under argon.

Thin steel, stainless and aluminum (aluminum needs AC). The neatest welds.

Every bit of skin covered; never thoriated (red) tungsten.



Bar and rod bender

Bends flat bar and round bar cold round a die, to the angle its stop is set at.

Brackets, hooks, hairpin legs, scrolls. Bend 2–5° past for springback.

Clamped down, fingers back from the die, clear of the handle's swing.



Tube bender `intermediate`

Bends round or square tube round a die its own size, without kinking it.

Curved legs, handles and hoops.

The right die for the tube; slow, steady pressure: a kinked tube can let go.



Box-and-pan brake `intermediate`

Folds sheet metal along a straight line, in a clamp, with a leaf you lift.

Brackets, channels, trays and covers from the flat pattern.

Within its gauge; fingers out of the clamping bar; gloves for the edges.



Snips and shear

Aviation snips cut sheet metal by hand; the foot shear cuts long, straight edges.

Cutting blanks out of a sheet; notching corners.

Sheet edges cut like a knife: gloves, and deburr every edge.